

# Experiment - 7

## 1) Aim/Objective:

Case Study of Amazon Web Services

## 2) Theory

Amazon Web Services (AWS) is the world's leading cloud computing platform, providing on-demand computing resources, storage, and a wide range of cloud services. Launched by Amazon in 2006, AWS revolutionized IT infrastructure by offering scalable and cost-effective solutions to businesses, startups, and governments.

AWS operates on a pay-as-you-go model, reducing the need for businesses to invest in expensive on-premise hardware. It provides over 200 fully featured services across computing, storage, networking, databases, artificial intelligence (AI), machine learning (ML), security, and the Internet of Things (IoT).

### ***AWS Architecture***

AWS is built on a robust and scalable architecture that ensures high availability and security. The key components of AWS architecture include:

#### **2.1 Regions and Availability Zones**

- AWS infrastructure is divided into **Regions**, which are geographically separated locations worldwide.
- Each Region contains multiple **Availability Zones (AZs)** to ensure redundancy and fault tolerance.
- Availability Zones are connected through low-latency networks to provide high availability and disaster recovery.

#### **2.2 Elastic Compute Cloud (EC2)**

- EC2 provides resizable virtual machines (instances) to run applications.
- It supports various instance types optimized for compute, memory, storage, or GPU-based workloads.
- Features **Auto Scaling** and **Elastic Load Balancing (ELB)** to manage traffic distribution.

#### **2.3 Amazon Simple Storage Service (S3)**

- S3 is an object storage service designed for scalability and data durability.
- Stores data in **buckets**, with versioning and lifecycle management capabilities.
- Ensures 99.999999999% durability (11 nines) and supports security features like **encryption** and **access control policies**.

## 2.4 AWS Identity and Access Management (IAM)

- IAM provides security controls for authentication and authorization.
- Uses **roles, policies, and multi-factor authentication (MFA)** to manage access to AWS services.

## 3. Case Study

One of the most impactful case studies of AWS adoption is **Netflix**, the world's largest online video streaming platform. Netflix transitioned from traditional data centers to AWS cloud to meet its **scalability, reliability, and global availability** requirements.

### 3.1 Background of Netflix

- Netflix serves **over 250 million subscribers** worldwide (as of 2025).
- It streams high-definition video content, requiring **massive storage, bandwidth, and low-latency content delivery**.
- Before cloud adoption, Netflix faced issues with **scaling infrastructure, downtime during peak demand, and global content delivery**.

### 3.2 AWS Services Used by Netflix

1. **Compute Services**
  - **Amazon EC2 (Elastic Compute Cloud)** for scalable virtual machines.
  - **Auto Scaling Groups** for handling fluctuating demand.
2. **Storage Services**
  - **Amazon S3 (Simple Storage Service)** to store movies, shows, and metadata securely.
  - **Amazon Glacier** for cost-effective archival storage.
3. **Content Delivery**
  - **Amazon CloudFront** for global content delivery with low latency.
4. **Data Analytics**
  - **Amazon EMR (Elastic MapReduce)** and **Amazon Redshift** for big data analysis.
  - **AWS Lambda** for serverless real-time video recommendations.
5. **Security & Compliance**
  - **AWS Identity and Access Management (IAM)** for secure access.
  - **KMS (Key Management Service)** for encryption of content.

### 3.3 Benefits Realized by Netflix

- **High Availability:** Netflix runs on multiple AWS regions to ensure **99.99% uptime**.
- **Global Scalability:** Seamlessly serves millions of simultaneous users worldwide.
- **Cost Efficiency:** Reduced capital expenditure by shifting from data centers to pay-as-you-go cloud.
- **Personalization:** Real-time AI-driven recommendations using AWS analytics services.
- **Disaster Recovery:** AWS supports multi-region failover and backup.

## ***4. Business Impact and Economics***

The AWS–Netflix case illustrates the **economic and strategic advantages** of cloud adoption:

- **Operational Efficiency:** Reduced overhead of managing data centers.
- **Faster Innovation:** Developers deploy and scale features quickly.
- **Global Market Reach:** Cloud-based content delivery enables serving customers in 190+ countries.
- **Customer Satisfaction:** Improved streaming speed, availability, and personalization.

## ***5. Challenges Faced***

While AWS provided immense benefits, Netflix also had to address:

- **Cost Management:** Optimizing resource usage to prevent high bills.
- **Data Privacy Regulations:** Complying with different country-specific data laws (e.g., GDPR).
- **Vendor Lock-in:** Heavy reliance on AWS ecosystem.

## ***6. Lessons Learned***

- Cloud adoption should align with **business scalability goals**.
- **Multi-region deployment** ensures high availability and disaster recovery.
- Cost monitoring tools (like AWS Cost Explorer) are essential.
- Cloud security must be managed under the **shared responsibility model**.

## ***7. Future Scope of AWS in Case Studies***

AWS continues to expand into:

- **Artificial Intelligence & Machine Learning (SageMaker, Bedrock)**
- **Serverless Computing (AWS Lambda, Fargate)**
- **Sustainability in Cloud (Carbon-neutral data centers)**
- **Quantum Computing (Amazon Braket)**